

Course Notes — SESSION 1

THE GENERAL PROBLEM OF INTELLIGENCE: CREDIT ASSIGNMENT

In the Course Introduction, we laid out a view of AI and organizations as direct analogs. **Intelligence** in either was seen as the **ability to learn to solve complex problems to pursue goals**. Before we start to explore this analogy, however, we need to agree on some ways of talking about what we actually mean by “learning” and “complex problems”. To do so, this week’s course notes provide a precise yet general way to talk about learning in AI and organizations called **credit assignment** that we can use throughout the semester. The notes also provide intuition on what we mean by a “complex” problem. We will unpack complexity more precisely in subsequent sessions.

Before we begin, you should note that current AI discourse you may be familiar with mostly talks about learning too narrowly for the purposes of our course — as a specific set of technologies called “machine learning”. Machine learning technologies typically learn **statistical patterns** from large volumes of data, for the purposes of making **statistical predictions**. Large language models (LLMs), for example, learn statistical patterns in text data to solve problems of predicting new text to generate. In organizations, these machine learning technologies are often viewed in narrow economic terms as “**prediction machines**” that **lower managers’ costs of prediction** (Aggrawal, Gans & Goldfarb 2018; 2024).

Most foundational AI ideas are about learning patterns other than statistical patterns. And their analogs in organizations are interpretive processes that go beyond statistical prediction. To explore the full range of foundational AI ideas and their relationship to organizations, we will have to step outside of the gravitational pull of the currently-popular “prediction machine” view of AI. We need a general way of talking about **learning *any* patterns** that might be helpful to solving complex problems intelligently.

Founders of AI referred to this general problem of intelligence as that of **learning to assign credit** (Samuel 1959; Minsky 1961; Holland 1975). Credit assignment refers to any process for **evaluating** (“assigning credit” to) **effects of individual actions** on solving a problem. In LLMs, individual actions are generating text fragments. Credit is assigned using statistical predictions to evaluate whether fragments are good.

During the course, we will look at many other actions and evaluation mechanisms beyond those considered in LLMs or machine learning. For this week, our purpose is to develop a nuanced understanding of the idea of credit assignment so that we have a shared, general way of talking about learning across diverse AI ideas and analogous phenomena in organizations. We will do so by looking at two paradigmatically-different learning strategies in AI and organizations, one called **reinforcement learning** and the other called **learning in a “society”**. Exploring and

debating tensions among these paradigms (or alternatives to either) as they relate to the general problem of credit assignment will be a focus of our journey throughout this course.

1. Introducing the credit assignment problem

To introduce the credit assignment problem, let us first consider a few common examples of learning to solve problems in organizations.

- (1) A new shipbuilding company is formed¹. It takes them a much longer time to build their first ship than their competitors, and the ship has many defects. They build their second ship, which takes them slightly less time and has fewer defects. Many years later, having built hundreds of ships, management claims to have “learned” to make them faster and with fewer defects than its competitors. Yet how did management know which changes to the vast number of interdependent actions in the shipbuilding factory actually *caused* faster production speed and fewer defects, versus which were merely *correlated*?
- (2) An app development team wants to improve the usability of a social media app². The team brainstorms a set of possible features that users might like. They evaluate each one using “A/B testing”, in which users are randomly given the app *with* a feature (version A) or *without* it (version B). By launch time, the team claims to have “learned” the optimal app features that its users want. Yet how could they tell whether they learned the features of a social media app that users *actually* want, or just which features users *relatively* want among the ones that they tested?
- (3) A budgeting committee³ for a pharmaceutical company meets at the beginning of every quarter to decide how to allocate the marketing budget across business units responsible for different drug categories. The committee allocates their budget based on information about each business unit’s sales for the previous quarter, as well as by talking to business unit heads and reading reports about market trends in the industry. After several years and many quarterly meetings, sales consistently increase. The committee claims to have “learned” how to allocate the budget effectively. Yet many business unit heads believe sales would have grown *even more* if more budget had been allocated to *their* units.

1.1 Individual actions as “generate-and-test” processes

What should you take away from these examples? For one, they embed some shared building blocks of individual actions that we can use for talking about credit assignment across any AI idea or organizational theory. As depicted in Figure 1 below, all three examples involved (1) **goal-directed agents** who (2) **generated actions** (3) **tested** the effects of actions on goals using (4) **trial-and-error feedback**, and used this feedback to (5) **modify actions**.

¹ cf. Argote & Eppler (1990).

² cf. Thomke (2020).

³ cf. Cyert & March (1963).

The agents were shipbuilding workers of managers, app team developers, or budget committee members; their goals were reducing costs and defects when building a ship, improving the features of a social media app, or increasing sales across business units. Actions generated included building ships, experimenting with app features, and allocating quarterly budgets. Testing meant using feedback to assign credit to the generated actions according to their effects on goals. Feedback included data on ship defects, on how users engaged with app features, or on business unit sales figures. Modifications of actions were refinements to shipbuilding techniques, adding or removing app features, or changes to business unit budgets.

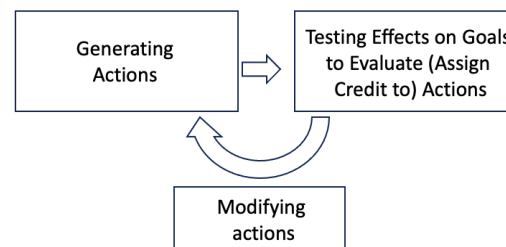


FIGURE 1:
Generating, testing, modifying actions to learn to solve problems

1.2 Credit assignment as the problem of generating intermediate feedback

The three examples also embed the concept of assigning credit — that of learning to evaluate the effects of individual actions on solving a problem — as one of effectively using feedback to modify subsequent actions. Testing the effects of any particular action (e.g., choices of shipbuilding techniques, app features, or business unit budgets) on a goal required understanding how it depended on other actions. We will refer to this requirement as the challenge of assigning credit arising from the **interdependence of actions**. Further, the examples all show how “tests” of actions do not necessarily provide adequate feedback for evaluating them. **Feedback** about goals (e.g., ship defects, user issues with the app, or changes in pharmaceutical sales) was either **delayed** until well after actions were generated, or **ambiguous**. While the organizations “learned” something through the process depicted in Figure 1, *there was no easy way to know whether they learned the right things* due to these challenges of assigning credit of *delayed* and *ambiguous feedback* regarding *interdependent actions*.

The shape of the **general solution** to challenges of **credit assignment** was initially described by the IBM scientist **Arthur Samuel** in 1959 regarding his pioneering AI program for learning to play the game of checkers. Good checkers players, Samuel noted, do not wait until the end of a game to assign credit to sequences of interdependent actions (moves). During a game, they count the number of pieces left, and monitor how the pieces are distributed across the board to evaluate how they are doing. The number and distribution of pieces are proxies indicating progress towards a goal. Samuel called the evaluation of such proxies the use of “**intermediate feedback**” to assign credit.

Solutions to the credit assignment problem, Samuel argued, are generally of this form: they are *ways of generating-and-testing various types of intermediate feedback*. As examples, the shipbuilding company may have counted the number of defects in particular parts; the app development team may have defined key performance indicators (KPIs) for each step of a customer “journey”; the budgeting team may have set milestones for business units and required periodic reports from them.

Samuel’s checkers program generated-and-tested various sources of intermediate feedback under one paradigm that has become known in the AI field as reinforcement learning. In a famous 1961 agenda-setting paper for the AI field, **Marvin Minsky**, co-founder of the MIT AI lab, disagreed that reinforcement or similar strategies offers a general solution to assigning credit. He proposed a distinct paradigm he called learning in a “society of mind”. Next, we explore these two paradigms, both to better understand the idea of credit assignment and to encourage you to start to think critically about the diverse types of solutions to the problem that may be possible.

2. Paradigm 1: assigning credit by reinforcement learning

In **reinforcement learning**, an agent assigns credit simply by “reinforcing” actions that seem to have the effect of getting it closer to its goal. Any such effects of actions on goals are called **rewards**. The overall goal in reinforcement learning is to “**maximize rewards**”. This view of an AI agent is similar to a utility-maximizing individual in microeconomics⁴. By seeing how the process of reinforcing and maximizing rewards operates, you will get a basic feel for the dynamics of credit assignment, and of how economic (as distinct from organizational) ideas may more generally run into limits as ways of generating intermediate feedback.

Let us first put the above description a bit more formally. **Reinforcement** means that, when an action leads to a reward, credit is assigned to this action by **increasing the probability** that it will be generated the next time. We will call these probabilities for generating actions “**beliefs**”, in that they capture not facts about reality per se, so much as an estimate by some agent of how much credit individual actions should get based on whatever intermediate feedback the agent has received so far. The amount of credit that the agent assigns to an action — the amount that the probabilities of generating an action are modified when a reward is received — is based on what is called the agent’s **policy**. Put another way, a policy⁵ in reinforcement learning is a set of rules defining how to update **probabilities** of beliefs about actions.

⁴ The similarity of rewards to microeconomic utility functions arises from later integration (from the late 1970s) into reinforcement learning models of econometric techniques from operations management (Sutton & Barto 2015, Ch. 1; Mirowski 2003). I will use the terms “reinforcement” and “rewards” here with some reluctance, in deference to their current familiarity. Samuel’s (1959) and earlier foundational studies in this area (e.g., Shannon 1950) make no mention of either term, and were based on electrical and software engineering models rather than econometrics.

⁵ A policy in reinforcement learning in AI is the same concept of a policy used in formal models of decision-making in fields such as political science, economics, and management theory.

Hence, learning to assign credit here refers to updating (i.e., modifying) the probabilities for generating actions as trial-and-error feedback accumulates. The policy refers to the amount of updating after each action, called a **learning rate**, that ranges from 0 to 1, and which is multiplied by the prior belief (probability) to get the new belief. This learning rate can be different for “successful” actions that increase rewards versus “unsuccessful” actions that decrease rewards⁶. Hence, in reinforcement learning, learning can be expressed as follows:

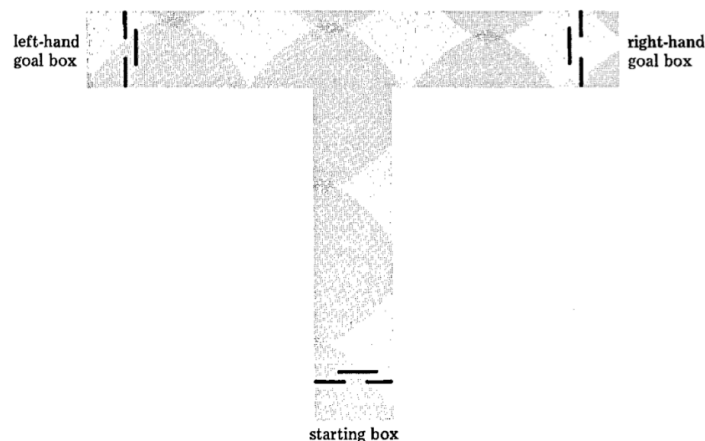
$$\text{Updated belief} = (\text{learning rate}_{\text{success}}) * (1 - \text{prior probability of generating action})$$

or:

$$\text{Updated belief} = (\text{learning rate}_{\text{failure}}) * (1 - \text{prior probability of generating action})$$

2.1 Navigating a T-maze

To concretize the above discussion, let us first look at the use of reinforcement to assign credit in a simple problem. In fact, let us look at perhaps the simplest type of reinforcement learning problem, in which the policy concerns how to assign credit among just two possible actions. Imagine a mouse who generates actions in a T-shaped “maze”⁷ (i.e., the simplest kind of maze we can think of) to find cheese as a reward, as depicted in Figure 2. The mouse is placed at the bottom of the maze, and looks for cheese once a day for 1,000 days, with a goal of maximizing the amount of cheese it eats. It can take only one of two possible actions, turn left or right. The left hand “goal-box” and the right hand “goal-box” of the T-maze are hermetically sealed, so that the mouse cannot get “feedback” from smelling the cheese until after generating a “left” or “right” action. Imagine that each day, a human manager of the maze decides where to place the cheese by flipping a coin, where Heads = Left, and Tails = Right. Further, imagine a coin that is weighted on one side so that the average number of “heads” for 1000 tosses will be 700 (70%) and the average number of “tails” will be 300 (30%).



⁶ A well-known finding in research on organizational learning is that failed trials are more likely to induce a change in actions, with successes more likely to encourage maintaining the status quo.

⁷ cf. Denrell, Fang & Levinthal (2004).

FIGURE 2: A T-maze (Lave & March 1975: p. 250).

The mouse’s “policy” should be to go left every time, as going left will always be more likely (70% versus 30%) to have cheese. Yet on Day 1, the mouse has no feedback to believe that left is different from right — that is, it has no basis to assign credit either to the action of going left or to the action of going right. *Its best policy is to act randomly — to go left 50% of the time and right 50% of the time.* Now, imagine the mouse randomly goes left and finds the cheese. What does it do on Day 2? The key idea is that, rather than suddenly go left 100% of the time, it realizes that the feedback from Day 1 is really only intermediate. That is, it has a credit assignment problem. It knows that the human maze manager may not make things so simple. Instead, our mouse designs a policy of using a **learning rate** that updates (increases or decreases) *probabilities* of generating a left or right action in the next trial.

To avoid updating probabilities beyond 100%, we need to assume that the mouse’s learning rate is only proportionate to whatever is left to learn (Lave & March 1975: 254). For example, if the initial beliefs (probabilities) are 50% left and 50% right, then we will say that the mouse has 50% more to learn. Under a 10% learning rate of going left, and assuming left is rewarded, then the probability for trial 2 will be updated by $0.5 + (0.1(1-0.5))$, or 55%, and the new belief will be 55%. As shown in Table 1 below, for both 10% or 30% learning rates, updates get progressively smaller as the amount left to learn is reduced. Note in both columns of Table 1 that, while the mouse’s policy still is not optimal after eight trials, it has at least “reinforced” actions that are closer. After 1,000 days of updating its beliefs through trial-and-error for either learning rate, the mouse will likely have enough feedback to have not just updated its beliefs, but to have “learned” a policy simply to go left 100% of the time, maximizing its reward of cheese.

Table 1

Trial	Probability of going left under a 10% learning rate (more “exploitative”)	Probability of going left under a 30% learning rate (more “explorative”)
1	0.5	0.5
2	0.55	0.65
3	0.595	0.755
4	0.636	0.829
5	0.682	0.880
6	0.714	0.916

7	0.742	0.941
8	0.768	0.959

Since the mouse seems to have the ability to learn to eventually solve the T-maze problem under either learning rate, should we call it “intelligent”? Let us leave aside whether a T-maze is too simple to be a true test of intelligence. Instead, the general way in reinforcement learning of thinking about this question is to start by assuming that **feedback is costly** (again, note the close ties here to economics — here, the scarce resource to be “economized” is that of doing a trial to get more feedback). Hence, intelligence requires policies that *balance the need to assign credit to beliefs efficiently, not just effectively*.

Compare, for example, how the learning rates in Table 1 change the efficiency and effectiveness of updating the mouse’s probability of going left. Under the 10% learning rate, even if credit is assigned ineffectively for any particular trial (i.e., the mouse gets a reward for going right), it will result in only small updates to the probabilities. This conservatism requires an *inefficiently large number of trials to assign credit effectively* (i.e., to identify the policy of going left 100% of the time that will maximize rewards). Conversely, under the 30% learning rate, if credit is assigned accurately initially, the mouse efficiently updates its beliefs rapidly in the right direction. This boldness, however, carries the risk that, if the wrong actions happen to be rewarded early on, then the mouse may assign credit ineffectively, and its beliefs may rapidly converge in the wrong direction. Alternatively, a high learning rate may lead its beliefs to fluctuate wildly and never converge on the policy that maximizes rewards.

AI and organizational researchers call a policy of using a low learning rate **exploitation**, in that it “exploits” learning based on prior trial-and-error actions (rather than rapidly “forgetting” this learning by making sudden large updates to beliefs). Conversely, they call a policy of using a high learning rate as **exploration**, in that making large updates amounts to taking a risk on a new belief without much basis in prior feedback, and then seeing if it works. The need to balance the efficiency and effectiveness of these policies is famously called the **exploration-exploitation tradeoff**. This tradeoff frames the solution to the fundamental problem of credit assignment in our first AI idea. In reinforcement learning, the solution is simply to *design a policy (learning rate) that balances the trial-and-error exploration and exploitation of beliefs*.

2.2 Navigating two, interdependent T-mazes

Next, imagine that our mouse instead has to navigate *two* T-mazes, as depicted in Figure 2 below, and where the cheese is only placed in the second one. The mouse now has to assign credit to two sets of **interdependent** beliefs instead of one. That is, it has to update probabilities for going left or right in the first T-maze *and* in the second T-maze. The key idea here is that the credit

assignment problem is not just twice as hard! The second T-maze in fact is itself *really two different ones*, depending on the **sequence of actions** taken in the first T-maze. By adding just a *single action* to our T-maze, we ended up with *four paths* of actions to which to assign credit. You should see that continuing to add a third and fourth T-maze would make things far more challenging still.

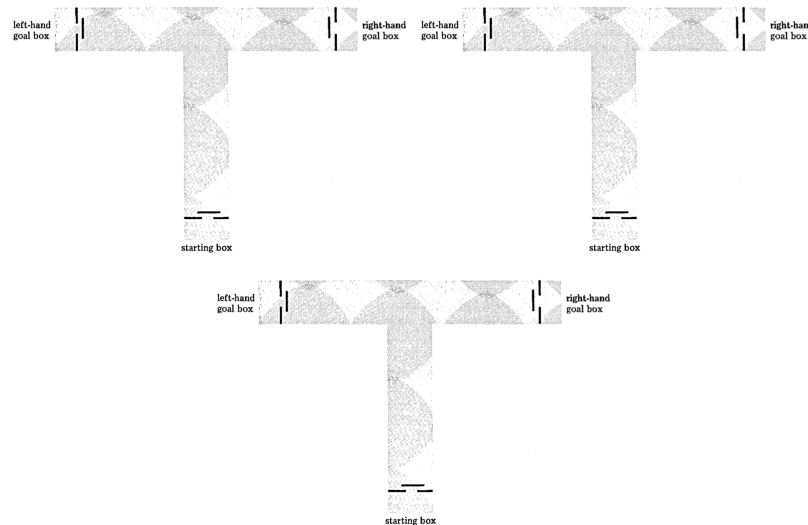


FIGURE 2: Two t-mazes increases complexity combinatorially

More precisely, *a linear increase in interdependent actions leads to a combinatorial increase in the cost of trial-and-error learning*. In fact, it can be formally shown that even small increases in the number of interdependent actions, measured as the **combinatorial complexity** of a problem, overwhelm the effectiveness of trial-and-error learning policies such as those in Table 1. This is called **combinatorial explosion**. Combinatorial explosion helps explain why our three organizations struggled to assign credit to what they had learned. Lacking enough intermediate feedback about interdependent actions, they could not tell if the beliefs they reinforced about these actions were true or just “superstitious”. This phenomenon — reinforcing beliefs without enough feedback to assign credit — is called **superstitious learning** (Levitt & March 1988).

2.3 Navigating arbitrarily complex mazes: Sutton’s “reward hypothesis”

In the late 1970s at the University of Massachusetts, Amherst, **Richard Sutton** and his colleague **Andrew Barto** developed a reinforcement learning approach to address combinatorial explosion that they called **temporal differencing**, or **TD**. Their key insight was that learning *may not in fact require beliefs about interdependent actions*. Instead, an agent may learn to act effectively by assigning credit to a much simpler “**value function**” that only estimates actions’ **aggregate rewards** (also called their “**cumulative expected value**”).

The insight is that, rather than rely on intermediate feedback in a given domain (such as counting checkers pieces in the domain of checkers), you start by just generating tentative beliefs about

the rewards of actions during intermediate steps. Then, after the actual feedback on the intermediate trials is received, you can just update the beliefs based on the evidence. The key idea is that, similar to the “time value of money” used in economic models for valuing the cash flows of investments, how much you update beliefs depends on the time difference (hence the name) between successive predictions.

The intuition is much like that of economic valuation techniques such as **net present value (NPV)** that managers commonly use when deciding to invest in a project. NPV estimates the “reward” of an investment simply as the sum of cash flows generated by all the complex actions taken during a project. Intermediate feedback is simply any cash flows observed during a project. The manager uses this feedback to modify the cumulative expected value of the project, *without needing to know which actions to assign credit to or “blame”*. Put another way, rather than reinforcing specific actions, the agent reinforces only the cumulative outcomes that interdependent actions generate. Sutton and Barto claim that this approach offers a general solution to the credit assignment problem, corresponding to what they call the “**reward hypothesis**”.

***REWARD HYPOTHESIS:** “all of what we mean by goals and purposes can be well thought of as the maximization of the expected value of the cumulative sum of a received scalar signal (called reward)”.*
(Sutton & Barto, p. 57).

The “reward hypothesis” suggests credit can be assigned just by evaluating how things are going and modifying actions in ways that seem more valuable. TD is their trick to do so. It refers to how any intermediate feedback during one time period can be used to modify one’s running estimate of the cumulative rewards at any “temporally-different” period as a basis for assigning credit to actions. To bring things back to our multiple T-maze example, TD allows the mouse to use the eventual discovery (or not) of cheese to estimate rewards at any of the earlier sections in the maze, without having to evaluate paths of interdependent actions. These reward estimates can then be used as a basis for evaluating and reinforcing individual actions (i.e., for solving the credit assignment problem).

3. Minsky and the limits of the reinforcement paradigm

Sutton’s reward hypothesis proposes reinforcement learning as a general solution to credit assignment for any problem requiring intelligence. Is this really plausible? A counterargument in the AI field is from Minsky. In his view, credit assignment is not about *finding solutions* to maximize rewards, nearly as much as it is about *understanding* problems themselves. As he emphasized, “*you first must be able to do something before you can be rewarded for doing it*”.

Minsky argued feedback is not just uncertain or ambiguous but **sparse**, meaning that only a small subset of actions generate feedback useful for assigning credit. Consider, for example, if our shipbuilding company experiences a massive and prolonged fall in sales, as competitors begin to produce ships with low costs and defects. Its managers realize they can no longer survive as a traditional shipbuilding company and must change their strategy. Yet they lack even superstitious beliefs for what novel strategies may work, and have no clear way to think of “rewards” — feedback is sparse regarding most of the possible actions they could take to reformulate their strategy.

How to learn to assign credit for such problems of sparse feedback? Minsky proposed that doing so would require at least two key departures from the reinforcement paradigm that we just looked at. First, feedback needs to be *actively constructed to understand the nuanced structure of the problem itself*. Whereas Sutton sought to *avoid* such structure (i.e., how beliefs are interdependent) by using a simple value function, Minsky went the opposite direction in seeking to assign credit not only to beliefs about actions, but to understandings of how actions fit together. Second, Minsky argued that, since problems of intelligence are complex, agents construct feedback about only small parts of it. Rather than Sutton’s idea of reducing problems to a single value function, learning to assign credit is done in what Minsky called a “**society**” (i.e., organization) of multiple agents that generate intermediate feedback based on **coordinating** and **cooperating** actions. We next explore this idea of **learning in a society** as an alternative paradigm for thinking about the general problem of credit assignment.

4. Paradigm 2: assigning credit by learning in a “society”

4.1 Revisiting the T-maze

To introduce this alternative paradigm of assigning credit by learning in a society, let us revisit our earlier T-maze problem from a different point of view. Imagine the problem is not for a mouse to find the cheese in the maze, but for a human — let us call this human Geoffrey — to construct (i.e., design and build) the maze itself. Geoffrey is provided with certain blocks of wood for building the walls of the maze, as well as some smaller blocks for building the starting box, left-hand goal box, and right-hand goal box in Figure 2. Minsky suggests we think of the fundamental actions here as **managerial** or **administrative**: they concern *delegating actions to*

other “agents”, rather than actual designing or building. Geoffrey is a sort of “CEO” agent, whose actions manage lower-level agents in his mind or body needed to construct the maze.

Let us say, for example, that Geoffrey delegates the overall problem of constructing a maze to a “Builder” agent in his mind, who is specialized in making stuff like mazes out of blocks. “Builder” does not actually know how to build a maze, only that certain other agents, including one called “Add” (who takes actions of adding blocks to the maze) is needed for the job. Yet “Add” itself does not actually know how to construct the walls. It does know, however, that it needs another agent “Grasp” to pick up the actual blocks, and “Move” to put them in the position of the maze. These agents, the story continues, only know that they need to delegate their actions to thousands of other agents who control the muscle fibers of Geoffrey’s fingers.

In this set up, how should we think about learning to assign credit? In some sense, each agent’s action of delegating to other agents might be thought of as a “policy”, just as in reinforcement learning. Geoffrey’s “policy” is to delegate to “Builder” when the problem is constructing a maze, perhaps having “learned” to assign credit to this agent from past experience of using it to build mazes successfully. Yet stop and think: what is the goal (i.e., “reward” to be maximized)? Can such goals really be allocated to each individual agent?

Certainly, Minsky pointed out, one goal has to be that the T-maze should have the property of **containment**, meaning that the mouse cannot escape the maze and only take left or right actions. Yet *who* is rewarded for this goal? Consider that “containment” is a property of the entire *maze*, not of any of the blocks. Hence, neither “Builder”, nor “Add”, “Remove”, or “Grasp” agents are really rewarded for this goal. So who is rewarded? Minsky’s answer is that it is *all of the agents together*, and what is “rewarded” is how they have structured into a *society* for constructing mazes.⁸

Could Geoffrey’s intelligence be explained as a “society” where agents in his mind individually learn to reinforce actions that maximize rewards for their own individual goals (as in a free market), such that the maze and its containment property just emerge? Minsky says no. For one, given a goal of “containment”, for Geoffrey and other higher-level agents (e.g., “Builder”) to delegate actions effectively to lower-level agents (e.g., “Builder” and “Grasp”), *they must have already had some theories that **decomposed** the overall problem into simpler subproblems in some meaningful way.*

That is, our “CEO” Geoffrey is really always also partly a “designer” or “architect”, who has some theory of how the agents’ actions will fit together to, e.g., achieve the “containment” property in the maze. Hence, learning to assign credit cannot be abstracted from the problem of

⁸ For Minsky, such a society of agents may exist within a single human mind (i.e., the different components or parts of the brain or nervous system), an AI system, an organization, or in a conventionally-understood society itself (cf. Wiener XX).

learning about the “society” itself — what its structure is and how it functions, and how this all maps to the structure of the underlying problems. To be intelligent, Geoffrey learns to assign credit not like in reinforcement learning to some beliefs about some interdependent set of actions, nor to a value function, but rather to some (1) *space of possible decompositions of a problem* and to (2) *ways of organizing actions for solving this decomposed problem into a society of agents*.

Learning here is not about exploring or exploiting specific actions, so much as considering how to use feedback to modify the overall decomposition of the problem and how actions for understanding it or solving it are delegated to diverse agents in the society. These agents need not “maximize” individual rewards (though this is one possibility); mostly, they fit into the society by learning to coordinate and cooperate with many other agents using a variety of evaluative criteria for assigning credit, always evaluated with respect to actions of many other agents.

4.2 Constructing two different mazes

You should start to see how, even for a seemingly simple problem of constructing a T-maze from a few blocks, the credit assignment challenge is radically different from that in the reinforcement learning example of the mouse in the T-maze. Next, consider how these differences also have distinct implications for our earlier discussion of complexity. In the mouse example, the challenge of assigning credit increased combinatorially as we added more actions, such as making a mouse go through two T-mazes instead of just one. Yet when learning in a “society” to construct two T-mazes, *such an increase in actions adds little to the credit assignment challenge if the actions are all the same*. The reason is that constructing two T-mazes should involve pretty much the same society of agents as constructing one, since there are no new actions. Once one has understood how to construct one T-maze, constructing another one is a snap!

In contrast, what if Geoffrey encounters, or imagines, a new problem of constructing a T-maze, out of rows of hedges, rather than blocks of wood? In this case, Geoffrey has to learn to assign credit to a “new” society of agents, with agents specialized in, for example, trimming hedges. Yet certainly, in this new hedge maze, the goal of “containment” is still maintained. It would be inefficient to throw away all the credit assigned when constructing the first T-maze from blocks of wood. Geoffrey’s credit assignment challenge is, thus, Minsky claims, largely a matter of *figuring out what to remember and what not to* about how the previous problem was decomposed and managed. Yet it is not only Geoffrey, but all the other agents (“Builder”, “Add”, etc.) who also have to figure out what to remember when constructing a new hedge maze. Hence, Geoffrey is really not just a CEO managing how to delegate actions for constructing a maze to lower-level agents, but also how to manage their memories that enable efficiently constructing new mazes.

4.3 The management hypothesis

Geoffrey's credit assignment challenge is vastly more complex (or, at the least, qualitatively different) from the challenge in the reinforcement learning paradigm of modifying beliefs or value functions. Actions to assign credit include not just going left or right in a maze, or even delegating subproblems for constructing a maze, but also actions for how and whether to store feedback as a memories for future use and when to use these memories. For a lower-level agent, such as "Grasp", such memory-related actions might concern just when to delegate grasping actions to the muscle fiber "agents" in Geoffrey's hands. At the higher-level of Geoffrey himself, memory-related actions amount to learning to store and use feedback about his theories or designs of the overall maze and how to organize its construction.

Finally, since there are likely far too many agents for Geoffrey to remember in detail, he also needs to simplify all these memories, such as decomposing agents into hierarchical levels and assigning credit to their actions at these aggregate levels, akin to managing an organization. All of this leads to an overall view of how to address the general problem of credit assignment that we will call in this course the "management hypothesis".

MANAGEMENT HYPOTHESIS: *All of what we mean by goals and purposes can be well thought of as managing processes of decomposing problems, and memories about these problems, into societies of agents, and assigning credit within these societies in ways that enable them to satisfy some higher-level evaluative functions.*

Managing processes of learning to assign credit in a "society" means that, when the society collectively generates actions that lead to "rewards" in a broader sense (i.e., either of some economic reward, or just understanding things better), what is "reinforced" is really what Minsky called a "*society of memories*" about some goals. Put another way, "reinforcement" here increases the probability that a *configuration of agents* forms again in similar circumstances. Yet this overstates the comparison between the "reinforcement" and "society" paradigms of learning to assign credit in AI. Minsky's societies are agents who operate at different levels of abstraction, with different goals, and different knowledge, rather than on a single value function or belief system. Further, when they collectively assign credit to their configuration of agents, the novel configuration may form a new goal. Even if there is just a rearrangement of goals, the "reward" is not maximization, so much as learning a new skill or satisfying the needs of the society.

5. Conclusion

What should you take away from our discussion so far? First, you should see that any AI ideas (or analogous organizational theories) can be understood in terms of how they propose solving the general problem of assigning credit. Second, you should see from our two discussions of the credit assignment problem (reinforcement learning, and learning in a society) that different AI ideas can offer *paradigmatically* different views of learning and intelligence. Hence, understanding AI will require you to form *your own perspective* based on thinking critically about diverse ideas about intelligence that are often in tension.

6. Appendix

6.1 Historical link between the AI ideas and organizations

Reinforcement learning has roots in 1940s-1950s research in the US, famously by **Richard Bellman**, an applied mathematician at a US government think tank called the **RAND Corporation** in Santa Monica, California. Called “operations research”, early computers were combined with econometric techniques to optimize inventory, logistics, and supply chain processes for US military and corporate bureaucracies that were rapidly growing and becoming more complex in the post-World War II era. The **logic was economic**: given a measurable “reward”, compute policies to maximize it. This logic carries forward today. Technology platforms rely on reinforcement to control how large volumes of data inform actions. Social media firms such as Facebook treat “engagement” (how much users interact with the platform) as a reward to be maximized, without explicit beliefs or theories of social interaction. Today, reinforcement is being applied to far broader domains, and for far more general rewards. Its proponents argue credit assignment by maximizing rewards should generalize to any problem.

The AI idea of learning in a society begins with Minsky’s work at MIT from the late 1950s. It indirectly relates to work during the 1940s-1950s of **Herbert Simon** (1947) at Carnegie-Mellon on how organizations represent problems and goals, and of **Norbert Wiener’s** (also from MIT) proposal that actual societies required a more complex mathematics to be managed effectively, which he called **cybernetics**. Minsky was not interested in optimizing operations in military or corporate hierarchies; he was asking how intelligence itself works. His answer, based in part on observations of children playing with blocks (called the “**blocks world**”, and used in our earlier example of building a T-maze), is fundamentally organizational. Though reinforcement and other machine learning strategies have dominated popular discourse about AI in recent decades, current uses of AI increasingly involve managing systems of AI “agents”. Hence, as AI becomes increasingly embedded in organizations, the AI field may be circling back to Minsky’s paradigm.

6.2 Philosophical background

6.3 Further reading

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